

Explaining public support for net-zero climate policy instruments: Perceptions of distributive fairness under competing frames

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ABSTRACT

Understanding public support for net-zero climate policy instruments is crucial for policy implementation and durability. Using survey data from a Canadian sample ($n = 2362$), we examine support for six net-zero policies, focusing on the roles of distributive fairness, effectiveness, and message framing. Consistent with prior research, we find that fairness perceptions are the strongest predictor of support, followed by effectiveness, though fairness judgments vary across policies. Notably, opposition to a zero-emission vehicle mandate and to a lesser extent an electric appliance mandate rivaled that of a consumer carbon tax, challenging assumptions that regulatory policies face less resistance. Distributive fairness perceptions were most influenced by expected impacts on future generations, low-income households, and rural communities, while those who prioritize equality and need-based justice principles were less likely to view policies as fair. Finally, pro-policy message frames did not shift policy support when positioned against a competing anti-policy frame, adding to the evidence that compelling counter arguments can neutralize otherwise persuasive frames. These findings highlight the need for policymakers to integrate fairness considerations into policy design and communication strategies to enhance the long-term feasibility of net-zero policy instruments.

1. Introduction

Numerous governments worldwide have targeted net-zero greenhouse gas (GHG) emissions by 2050 to mitigate climate change. Achieving these targets requires selecting policy instruments that are both effective and politically viable. Public support is a key consideration in policy design, as it can help secure the political mandate necessary for implementation. While public support does not always translate directly into policy implementation, understanding its determinants can inform strategies to enhance political feasibility (Kallbekken, 2023).

A substantial body of research has identified multiple factors shaping support for climate change mitigation policies, including climate change concern, psychological and demographic characteristics, and policy-specific beliefs (Bergquist et al., 2022; Heyen and Wicki, 2024; Maestre-Andrés et al., 2019). Among these factors, perceptions of fairness and effectiveness are particularly influential—policies perceived as fair and capable of reducing emissions tend to receive stronger public backing (Bergquist et al., 2022; Maestre-Andrés et al., 2021). Fairness judgements, in turn, can be shaped by whether individuals believe a

policy benefits or burdens certain social groups (Nelson and Kinder, 1996). Social identity theory suggests that fairness perceptions can be group-centric, meaning that people evaluate policies based on their perceived effects on groups they identify with or see as politically relevant (Huber, 2022; Tajfel and Turner, 2004). While fairness is a strong predictor of support, less is known about how perceptions of group-specific impacts shape individuals' fairness assessments with respect to climate policy (Huber, 2022).

Additionally, another potential mechanism for influencing public perceptions is message emphasis framing—a communication strategy that highlights specific aspects of a policy to shape public interpretation (Druckman, 2001). Research around framing effects on climate policy support has yielded mixed results, with variability across experimental contexts and over time (Fesenfeld et al., 2024). Notably, most studies employ one-sided frames, presenting only a single perspective on an issue, whereas real-world public discourse often involves competing (two-sided) frames (Chong and Druckman, 2007a). Given that public opinion is often shaped by exposure to competing messages, more research is needed to assess how these interactions influence policy support (Heyen and Wicki, 2024).

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To contribute to this literature, we aim to answer the following research questions.

- (1) What are levels of support for different net-zero climate policies?
- (2) How do perceptions of distributive fairness and effectiveness at reducing emissions influence levels of policy support?
- (3) How do competing message frames shape public support and policy-specific beliefs?
- (4) How do perceptions of policy impacts on specific social groups influence distributive fairness judgements?

Using survey data from a web-based sample of Canadian adults ($n = 2362$), we explore these questions in relation to six net-zero climate policy instruments. We assess the role of distributive fairness and effectiveness perceptions in shaping support, investigate how competing frames influence these perceptions, and explore how fairness judgements relate to perceived policy impacts on different social groups.

The remainder of the paper is structured as follows: Section 2 provides background on Canada's energy-economy and policy landscape, public support for climate policy instruments, and message framing, then lays out the conceptual framework guiding this study. Section 3 outlines our methodology including survey design and analytical approach. Section 4 presents and discusses our results, and Section 5 concludes with policy implications.

2. Background

2.1. Canada's energy-economy

Canada has an energy-intensive economy and ranks among the highest global emitters of GHGs, with oil and gas extraction, transportation, and electricity generation the largest contributors (Environment and Climate Change Canada, 2024; Harrison, 2023). Oil and gas production, primarily concentrated in Western Canada, accounts for a quarter of national emissions, while transportation emissions are concentrated in urban centres, especially in the populous provinces of Quebec and Ontario. Electricity generation varies considerably across provinces, with hydro-dominated provinces having lower emissions than those reliant on fossil fuels (Environment and Climate Change Canada, 2024). Past emissions reductions have largely resulted from fuel switching and efficiency improvements. The coal phase-out instituted by Ontario in 2014 remains the most significant action, cutting national emissions by 34 megatonnes annually (Rosenbloom, 2018). However, these reductions have been offset by rising emissions from oil sands production and transport (Environment and Climate Change Canada, 2024).

Canada's energy and climate policy landscape reflects its federated governance structure, with provinces controlling natural resources, electricity generation, and land use. Federal government measures include carbon pricing, clean fuel regulations, and methane regulations for oil and gas producers, but implementation varies across jurisdictions. British Columbia introduced North America's first broad-based carbon tax in 2008, while Quebec operates a cap-and-trade system. Federal 'backstop' policies have encountered resistance, particularly from resource-dependent provinces, which continue to mount legal challenges against them. Despite broad public support for climate action in principle, acceptance of specific policies—especially carbon pricing—remains contentious (Lachapelle and Borick, 2022; Thomas et al., 2022).

2.2. Public support for different types of policy instruments

Recent reviews and meta-analyses synthesize a large body of research on public support for different types of climate policy instruments (Bergquist et al., 2022; Dechezleprêtre et al., 2022; Drews & van den Bergh, 2016; Kallbekken, 2023; Maestre-Andrés et al., 2019).

These instruments are commonly categorized along a continuum of compulsion, from voluntary 'pull' measures (e.g., subsidies, public awareness campaigns) to compulsory 'push' measures, which include regulations and carbon pricing (Howlett, 2019). While pull measures typically receive higher levels of public support—particularly when they benefit consumers—recent research highlights their drawbacks, including concerns over cost-effectiveness, fiscal sustainability, and the political difficulty of phasing them out once implemented (Dua, 2024). Meanwhile, push measures tend to receive lower support, especially when they directly target consumers rather than producers. There is also mixed evidence regarding the public acceptability of alternative push instruments (Heyen and Wicki, 2024).

A wide range of factors influence public support for climate policies, including psychological characteristics, attitudes, and beliefs, as well as contextual and sociodemographic variables (Grelle and Hofmann, 2024). Additionally, various policy-specific attributes have been associated with increased public support, such as intrusiveness (Huber et al., 2020) and transparency (Grelle and Hofmann, 2024). Notably, Bergquist et al. (2022) identified perceived fairness—the extent to which people, a process, or a distribution, are treated or implemented equally or according to a criterion (e.g., need, merit) (Maestre-Andrés et al., 2019)—as the strongest predictor of policy support ($r = 0.65$) among the 15 factors examined in their meta-analysis. Fairness perceptions in climate policy evaluation are often categorized into distributive and procedural dimensions (Tyler, 2000; Klinsky and Dowlatabadi, 2009). Distributive fairness concerns the perceived equity of policy outcomes, whereas procedural fairness relates to transparency and inclusion in decision-making processes (Tyler and Lind, 1992). Maestre-Andrés et al. (2019) emphasized that fairness perceptions around climate policy are conceptually complex both in theory and measurement (Heyen, 2023) and found that distributive fairness was a stronger predictor of support than procedural fairness.

Moreover, Bergquist et al. (2022) found that, second only to perceptions of fairness, perceptions of a policy's effectiveness—the belief that a policy can fulfil a specific aim—is a strong predictor of support ($r = 0.54$). Further, the authors found a stronger relationship for "taxes and laws aimed at mitigating climate change ($r = 0.50$) than those aimed at changing behaviour" (p. 236).

In Canada, studies have identified various individual and contextual factors influencing policy support. Long et al. (2020) found that support for transport regulations was most strongly linked to their respondents' values and trust in government. Similarly, Rhodes et al. (2017) highlighted climate concern and trust in scientists as relevant factors. While policy knowledge does not necessarily increase support, information provision boosts support for regulations while reducing it for carbon taxes (Lokshin et al., 2024; Rhodes et al., 2014). Odland et al. (2023) found that support for home decarbonization policies is consistently associated with altruistic values, climate concern, trust in scientists, and higher education. Focusing on the role of trust, Kitt et al. (2021) emphasized government competence as a key factor across policies. Additionally, publicly available polling has tracked public perceptions of carbon pricing, home decarbonization, and public transit (Re.Climate, 2024).

Despite a growing body of Canadian public opinion research, the main gaps, compared to work based elsewhere, are that net-zero policy instruments are underexamined, and that most studies correlate general individual and contextual characteristics with policy support rather than examining policy-specific attributes.

2.3. Message emphasis framing

Message emphasis framing (or simply framing) is a key concept in communication theory. It refers to the strategic presentation of information to highlight particular aspects of an issue, intended to influence how audiences perceive and respond to it (Chong and Druckman, 2007b). In theory, framing can influence public opinion by making

certain considerations more salient, guiding individuals to evaluate policies as they relate to underlying values, worldviews, and political identities, which shape how people interpret and respond to messages (Druckman, 2001). For instance, emphasizing the environmental benefits of carbon pricing may appeal to audiences who prioritize ecological concerns, while economic frames, such as highlighting job creation and clean energy investments, may resonate more with those who prioritize the economy. Importantly, researchers have long stressed that framing effects depend considerably on contextual factors, meaning that it is not reasonable to expect specific frames to work all the time or across different audiences (Chong and Druckman, 2007b).

Many studies have investigated the effects of framing on climate and energy policy support, and the evidence is mixed (Heyen and Wicki, 2024). For instance, the influence on support of frames that emphasize the health benefits of climate policies varies substantially (Dasandi et al., 2022; Poortinga et al., 2023). Empirical evidence on framing effects is often produced by randomly assigning study participants to control or treatment groups. In a review, Fesenfeld et al. (2024) found that most studies employ one-sided frames—those that emphasize only one side of an argument around an issue. In contrast, there have been far fewer studies that use competing frames, which could better simulate how political elites on opposing sides of an issue often challenge each other's rationales (Chong and Druckman, 2007a). The comparative sparseness of competing-frame studies may reduce the relevance of research findings for real-world communication, where counter-framing and competing arguments are common. For example, employing competing frames around proposed clean energy in a survey of Americans, Aklin and Urpelainen (2013) found that they undermined the effect of otherwise effective one-sided frames.

2.4. Conceptual framework

2.4.1. Model of policy support

Our analysis employs a multivariate model integrating individual and contextual factors, policy-specific factors, and framing effects to explain public support for six net-zero climate policies. In this study, *net-zero* policies are defined as those designed to reduce GHG emissions to near zero by 2050. These policies span different sectors and regulatory approaches, reflecting real-world efforts to achieve long-term emissions reductions. However, in the survey, we described each policy in terms of its expected impact by 2030. This choice was made to enhance respondent comprehension, as 2030 represents a more immediate and tangible policy horizon while still aligning with interim emissions reduction goals. Thus, while the study assesses public support for policies with net-zero ambitions, the descriptions used in the survey focus on near-term policy milestones rather than the full trajectory to 2050.

Specifically, we examine six net-zero climate policies in the Canadian context, including existing, prospective, and hypothetical instruments. These policy instruments and their associated stringency levels were selected to be broadly comparable in ambition, reflecting their potential to contribute meaningfully to Canada's net-zero emissions goals. This comparability was informed by recent studies using energy systems modelling and policy scenario design in Canada (Canadian Institute for Climate Choices, 2021; Environment and Climate Change Canada, 2022; Hoyle et al., 2024; Navius Research, 2023). While the policies differ in design and implementation pathways, the levels presented here represent a stylized increase in policy ambition aligned with net-zero

planning assumptions. At the time of the survey (August 2024), these policies varied in their implementation status:

- consumer carbon tax¹ (nationally implemented);
- zero-emission vehicle (ZEV) mandate (implemented provincially in British Columbia and Quebec, set for national rollout in 2026);
- low-carbon fuel standard (nationally implemented);
- electric building equipment mandate (hypothetical, modeled on a British Columbia proposal);
- oil and gas GHG emissions cap (under regulatory review), and;
- emissions trading system for buildings and transport (hypothetical, modeled on the European Union's ETS2).

This approach aligns with previous climate policy support studies (Grelle and Hofmann, 2024) and is grounded in findings from Bergquist et al. (2022), whose meta-analysis identified key determinants of policy acceptance. The model focuses on two policy-specific factors: (1) distributive fairness—perceptions of how fairly policy impacts are distributed, and (2) effectiveness—the perceived ability of a policy to reduce emissions. In addition to these factors, Bergquist et al. (2022) showed that several individual- and contextual-level factors have consistently been found to have medium to strong correlations with climate policy support (i.e., irrespective of policy type). Therefore, we include the following factors as covariates but do not focus on them much in our analysis: climate change concern (e.g., Clark et al., 2003), trust in implementing institutions (e.g., Kitt et al., 2021), political ideology (e.g., Harring et al., 2017), self-transcendent and self-enhancement values (e.g., Dietz et al., 2005), level of education (e.g., Czarnek et al., 2021), and age (e.g., Dietz et al., 2007). Notably, many potentially relevant covariates, such as second-order beliefs about others' support (Drews et al., 2022), were not included in this model and warrant attention.

Additionally, we examine the influence of competing message frames on policy support and fairness/effectiveness perceptions. We used two main criteria to guide our selection of frames. First, we mirrored arguments that are commonly used in public debate in Canada regarding energy and climate policy. Second, we aimed to reflect frames used in the existing academic literature, which we take to be an indicator of what types of arguments might be important in terms of shaping public opinion—Section 3.1. provides wording and elaborates the related procedure. Our first pro-climate policy frame emphasizes an intergenerational fairness narrative. Here we follow Potential Energy (2023), who found that in many countries, including Canada, framing climate policy implementation as protecting the planet for the next generation substantially increased support for a variety of climate policies, even when positioned against competing anti-climate action arguments.

Our second pro-policy frame emphasizes the economic benefits of climate policy implementation, like many other studies. For example, Badullovich et al. (2020) found in their review of framing in climate change communication literature that a pro-climate action economic frame is one of the most common, while Li and Su (2018) also showed that economic factors such as job creation and energy costs, play a key role in discussions surrounding renewable energy. Moreover, Stoddart et al. (2016) found in their review of Canadian national media coverage of climate change that economic and energy interests were a dominant frame from the period 1997–2010, and Martel-Morin and Lachapelle (2022) demonstrated that a relative price frame increased policy support

¹ The carbon tax applied in this study was designed to represent the Canadian government's consumer-facing carbon pricing system, or its provincial counterparts, which applied to fossil fuel purchases. This reflected how the policy was commonly understood in Canadian public discourse. Note that Canada also has in place separate carbon pricing systems for large industrial emitters (e.g., the federal Output-Based Pricing System and various provincial programs), which are distinct from the consumer carbon tax examined here.

among audiences that were less engaged on climate change.

Drawing from the same literature, our anti-policy frame, which we position experimentally in opposition to both pro-policy frames, emphasizes the economic costs of climate action. While the majority of framing studies have focused on pro-environmental efforts, several studies highlight the use of similar economic-costs frames (Aklin and Urpelainen, 2013; Ansolabehere and Konisky, 2014; Bayulgen and Benegal, 2019; Potential Energy, 2023). For instance, Millar et al. (2021) showed that opponents of the province of Ontario's feed-in-tariff and cap-and-trade policies successfully framed them as measures likely to lead to increasing energy prices. We believe that an economic-costs frame reflects the contemporary elite discourse in Canada, although we have not conducted formal media analysis to be able to say for certain.

The following equation summarizes our model of policy support: $Policy\ Support_{ij} = B_{0i} + B_{1i}Distributive\ Fairness_{ij} + B_{2i}Effectiveness_{ij} + B_{3i}Frame_j + \sum_{k=4}^9 B_{ki}Covariates_{kj} + \epsilon_{ij}$, where i denotes policy-specific factors and j denotes individual and contextual factors.

2.4.2. Model of distributive fairness

To examine perceptions of distributive fairness as a predictor of policy support, we use a supplementary conceptual model wherein an individual's perception of distributive fairness is predicted by: (a) their perceptions of a policy's impact on different groups, and (b) their adherence to distributive justice principles. This model centres the observation that fairness judgements rely on which distributive justice principle or norm one applies in each context; measures may be legitimately viewed as fair by some and unfair by others, even given identical perceptions of the distribution of its goods and burdens. We rely on Adriaans and Fourré (2022) for definitions of four basic justice principles.

- (1) Equality stands for an equal distribution of rewards and burdens irrespective of individual characteristics and contributions.
- (2) Equity suggests a distribution based on merit through the basis of proportionality between individual inputs and involvement on one side and the profits to be enjoyed on the other.
- (3) The principle of need calls for a distribution that takes individual basic needs into account.
- (4) The entitlement principle suggests that ascribed (e.g., gender) and achieved status characteristics (e.g., occupational prestige) should determine the distribution of goods and burdens. (p. 2)

Moreover, we draw on the literature regarding group-centric thinking, which observes that prominent social groups often act as reference points for public opinion formation (Nelson and Kinder, 1996). Social groups can be understood as collections of individuals who share at least one politically or socially significant characteristic (e.g., profession, ethnicity, religion, social class) and identify as members of the group based on that shared attribute (Huber, 2022). According to social identity theory, people develop emotional attachments to certain groups in society, fostering positive feelings toward their 'in-groups' and negative perceptions of 'out-groups' (Tajfel and Turner, 2001, 2004). Thus, people can make judgements about policy options (consciously or unconsciously) by symbolically associating values with specific groups of people (Huber and Haselmayer, 2024). For instance, if someone does not pay attention to climate policy or politics, they may identify groups of supporters (e.g., environmental organizations) and opponents (e.g., outspoken politicians) to situate their interests—a type of decision-making heuristic (Kahneman, 2011; Simon, 1955).

We explore this concept in relation to distributional fairness by combining (a) measures of perceived policy impacts for each of nine different social groups (Konc et al., 2022), and (b) measures of a respondents' adherence to each of the four justice principles defined above. We also include the framing treatments in the analysis, and use

the following groups: future generations of Canadians, owners of small businesses, people who depend on clean energy jobs, people who depend on fossil fuel jobs, people with low incomes, people with high incomes, people who live in cities, people who live in rural areas or small towns, and shareholders of oil and gas companies. The following equation summarizes the approach: $Distributive\ Fairness = B_0 + B_1Frame + B_2Equality + B_3Equity + B_4Need + B_5Entitlement + \sum_{k=6}^{14} B_kPolicy\ Impact_k + \epsilon$

3. Methodology and data

3.1. Data collection and survey design

We designed a 15-min online questionnaire and implemented it with a sample of Canadian adults aged 19 years or older in August 2024. We hired a market research company, Leger Opinion, to recruit respondents by randomly sampling from qualifying members of their ~300,000-person internet panel of Canadians. A total of 2409 respondents completed the survey, of which 50 were removed for failing a quality control check after data collection (e.g., 'straight-liners' were removed). For the policy support models, the sample size was further reduced to 2312 after several respondents were removed due to missing values across predictors (e.g., level of education). Table 1 shows proportions of demographic categories of the sample in comparison to those measured by the 2021 Canadian Census. Notably, the sample was more educated and slightly younger, as is common in web surveys (Dillman et al., 2014).

The survey was made up of two main parts, corresponding to measurements of (a) individual and contextual factors, and (b) policy-specific factors and framing effects (see the full survey questionnaire in Appendix A). First, after several warm-up questions, we collected demographic information including age and level of education. We then estimated various attitudinal variables following methods used in other similar studies (Bergquist et al., 2022; Bumann, 2021). An indicator of trust in implementing institutions averaged respondents' trust of Canada's federal and provincial/territorial governments, legal system, and

Table 1

Proportions (%) of sample demographic characteristics compared to population statistics. Note that in the sample, the proportions given for Men+ and Women+ do not sum to 100 because we included other gender identities.

Characteristic	Sample	Population (2021 Census)
<u>Age</u>		
19-34	27.5	25.7
35-54	32.3	32.6
55+	40.1	41.7
<u>Gender</u>		
Men+	48.8	49.3
Women+	50.2	50.7
<u>Province/Territory</u>		
British Columbia	14.3	13.2
Alberta	11.9	11.6
Saskatchewan	2.9	3.1
Manitoba	3.8	3.6
Ontario	37.3	38.3
Quebec	23.6	23.2
Atlantic	6.1	6.6
Territories	0.0	0.3
<u>Education</u>		
University degree	40.2	26.7
No university degree	59.8	73.3
<u>Income</u>		
<\$50,000	27.5	26.9
\$50,000 - <\$100,000	35.7	32.6
\$100,000 - <\$150,000	20.5	20.3
\$150,000 - <\$200,000	10.5	10.3
>\$200,000	5.9	9.8

politicians (Kitt et al., 2021). Self-transcendent and self-enhancement value measurements were constructed by averaging respondents' adherence to the following 'guiding principles,' respectively: equality and equal opportunity for all, social justice, unity with nature, and a world at peace, and; influence and having impact of people and events, authority and the right to lead or command, and wealth and material possessions (Dietz et al., 2005; Odland et al., 2023). Respondents' levels of climate change concern were measured by asking respondents how worried they were on a five-point scale from (1) "not at all worried" to (5) "extremely worried" (Chryst et al., 2018). Political ideology was measured following a commonly used five-point left-right scale (e.g., Martel-Morin and Lachapelle, 2022). Finally, respondents' social justice orientations were collected via a series of four questions developed and tested by Adriaans and Fourré (2022), which measure respondents' adherence to four social justice principles—equality, need, equity, and entitlement (Van Hootegeem et al., 2021).

A second set of questions measured respondents' levels of policy support and perceptions of distributive fairness and effectiveness subject to differing framing treatments. Respondents were first presented with a

short summary of the issue of climate change mitigation, followed by the policy descriptions given in Table 2. Each respondent had an equal chance of being assigned to one of three framing treatment groups. The control group received only the policy definition, the first treatment group received the policy definition plus the 'intergenerational fairness' pro-policy frame along with the 'economic costs' anti-policy frame, and the second treatment group received the policy definition plus the 'economic benefits' pro-policy frame along with the 'economic costs' anti-policy frame. The policy information focused on achieving a policy-specific objective by 2030. The order of the policy instruments was also randomly assigned.

Next, each respondent, and for each policy instrument, answered a sequence of four questions, given in Table 3. These questions were our measures, in corresponding order, of policy support levels, perceptions of a policy's impact on different groups, perceptions of a policy's distributive fairness, and perceptions of a policy's effectiveness at GHG abatement.

Table 2
Policy definitions and frames.

Policy Instrument	Policy Information	Treatment 1	Treatment 2
Carbon Tax ^a	Governments can reduce emissions by making fossil fuels more expensive. To do so, they can add a fee to gasoline, diesel, home heating oil, and natural gas, then give money back to households to offset higher fuel prices. The fee usually goes up over time, and the money that people get back goes up too.	Some say that although putting a price on pollution will increase costs for people that use more fossil fuels, we need to act now to protect future generations. Others say that this carbon tax will raise everyday costs for driving and heating, making it hard for people already struggling financially.	Some say that putting a price on pollution is better for most households because they get back what they pay, plus they can save money long term if they use less fossil fuels or switch to clean technology. Others say that this carbon tax will raise everyday costs for driving and heating, making it hard for people already struggling financially.
ZEV Mandate	Governments can reduce emissions from transportation by requiring vehicle manufacturers to sell more electric cars and trucks each year, and slowly decrease sales of gasoline or diesel vehicles.	Some say that even though people might not want to switch to electric vehicles, we need to start now so that future generations aren't harmed by pollution from gasoline and diesel. Others say that it'll make things more expensive for manufacturers and households. They worry it'll cause job losses and make traditional gasoline and diesel vehicles more expensive.	Some say that this can boost our economy by encouraging innovation in the auto industry and creating new jobs in clean energy. Others say that it'll make things more expensive for manufacturers and households. They worry it'll cause job losses and make traditional gasoline and diesel vehicles more expensive.
Low Carbon Fuel Standard	Governments can reduce emissions from vehicles by requiring the suppliers of gasoline and diesel to make their fuels cleaner over time. To comply, fuel suppliers can invest in making their operations more energy efficient, blend biofuels into the gasoline and diesel that they sell, or pay others who are supplying electricity to electric vehicles.	Some say that although this policy might impose costs on some fuel suppliers as they adopt cleaner practices, it's worth it to ensure future generations are better off. Others say that this policy will bring extra costs for both fuel suppliers and consumers. They worry it'll raise prices for gasoline and diesel, making it harder for people to afford them.	Some say that this policy can help our economy grow by encouraging new ideas in clean fuel technology, as well as creating jobs. Others say that this policy will bring extra costs for both fuel suppliers and consumers. They worry it'll raise prices for gasoline and diesel, making it harder for people to afford them.
Electric Appliance Mandate	Governments can reduce emissions from buildings by requiring the suppliers of appliances used for things like heating and cooking in homes and small businesses to sell only electric appliances. For example, instead of selling furnaces that run on natural gas or heating oil, suppliers would be required to sell electric heat pumps.	Some say that even though clean appliances might add some short-term costs, it's important to reduce pollution now because it protects the well-being of future generations. Others say that banning fossil fuel appliances will make things more expensive for households and small businesses, adding to affordability concerns.	Some say that this can reduce energy costs and encourage innovation in energy-efficient technologies like electric heat pumps, which creates new job opportunities in the clean energy sector. Others say that banning fossil fuel appliances will make things more expensive for households and small businesses, adding to affordability concerns.
Oil and Gas Cap	Governments can reduce emissions by limiting the emissions that oil and natural gas producers are allowed to release into the atmosphere. To follow this rule, producers typically invest in clean technology, produce less oil and/or gas, or pay others who are investing in clean technology.	Some say that although oil and gas producers might incur costs from reducing pollution, it's worth it to protect the wellbeing of future generations. Others say that forcing rules on oil and gas producers hurts their bottom line, which means less production leading to job cuts, higher prices for consumers, and a less stable economy.	Some say that this can be good for the economy because it rewards innovation, creates jobs and encourages investment in clean technology. Others say that forcing rules on oil and gas producers hurts their bottom line, which means less production leading to job cuts, higher prices for consumers, and a less stable economy.
Buildings/Transport Carbon Market	Governments can reduce emissions from vehicles and buildings by requiring the suppliers of gasoline, diesel, natural gas, and home heating oil to have permits for the emissions that their fuels release when burned. Typically, fuel suppliers comply by investing in cleaner practices or buying permits from others who do.	Some say that while this might be costly for fuel suppliers, it's needed because reducing pollution now will make sure the next generation will have a better future. Others say that imposing extra costs on fuel suppliers risks economic instability, and that higher costs will get passed on to consumers, making driving and home heating more expensive for everyone.	Some say that this can drive economic growth by incentivizing clean technology innovation and energy efficiency in industry. Others say that imposing extra costs on fuel suppliers risks economic instability, and that higher costs will get passed on to consumers, making driving and home heating more expensive for everyone.

^a While broader GHG pricing instruments exist, this description reflects the design of Canada's consumer carbon tax, which applied to fossil fuel purchases by households and small businesses at the time of the survey.

Table 3

Measures of policy support level, perceptions of policy impacts on different groups, distributive fairness, effectiveness.

Variable	Policy Instrument	Question	Measurement
Policy Support	Carbon tax	Imagine that the fee goes up to 40¢ per litre of gasoline by 2030 and households are getting back about \$200 per month. To what extent would you oppose or support this policy?	• Strongly oppose • Oppose
	ZEV mandate	Imagine that by 2030, 6 out of every 10 new vehicles sold must be electric. To what extent would you oppose or support this policy?	• Neither oppose nor support • Support • Strongly support
	Low Carbon Fuel Standard	Imagine that fuels are required to have 20 % less carbon in them by 2030 compared to today. To what extent would you oppose or support this policy?	
	Electric Appliance Mandate	Imagine that by 2030, most new sales of appliances would be required to be electric, meaning that most new fossil fuel appliances wouldn't be allowed. To what extent would you oppose or support this policy?	
	Oil and Gas Cap	Imagine that by 2030, oil and gas producers must lower their emissions by 30 % compared to today. To what extent would you oppose or support this policy?	
	Buildings/Transport Carbon Market	Imagine that by 2030, total emissions that can be released from fuels used in buildings and vehicles are reduced by 20 % compared to today. To what extent would you oppose or support this policy?	
Policy Impact on Groups	All	This policy could affect different groups of people differently. In Canada, how much worse or better off do you think this policy would make ... - Future generations of Canadians - Owners of small businesses - People who depend on clean energy jobs - People who depend on fossil fuel jobs - People with low incomes - People with high incomes - People who live in cities - People who live in rural areas or small towns - Shareholders of oil and gas companies	• Much worse off • Worse off • Neither worse nor better off • Better off • Much better off
	All	Please indicate to what extent you agree with the following statement, where 1 is completely disagree and 5 is completely agree: Overall, the way that this policy affects these groups of people is fair.	• 1 Completely disagree • 2 • 3 • 4 • 5 Completely agree
Distributive Fairness	All		
Effectiveness	All	Please indicate how effective you think this policy would be at reducing emissions, where 1 means not at all effective and 5 means extremely effective?	• 1 Not at all effective • 2 • 3 • 4 • 5 Extremely effective

3.2. Data analysis

We relied on two different methodological approaches. We first conducted a descriptive analysis of respondents' levels of support, in addition to perceptions of distributive fairness and effectiveness at GHG abatement for the six policy instruments listed in Table 3. Then we explored factors of policy support and distributive fairness using a Bayesian sparse modelling approach called LASSOplus, introduced and documented by Ratkovic and Tingley (2017). The technique is designed to zero out the coefficients of all but the most relevant independent variables in its effects estimation routine, which mitigates overfitting and reduces the likelihood of omitted interaction bias. The technique also provides uncertainty estimates for its parameters. In our context, analysis using LASSOplus was recommended by Fesenfeld et al. (2024) in their review and re-analysis of framing effects on climate and environmental policy support—by comparison to ordinary least squares regression the method was shown to improve the quality of analysis with respect to treatment effects.

Of the nine independent variables included in our policy support models, we treated most as continuous, except for age groups, level of education, and the message framing treatments, which were specified as categorical variables. The dependent variable, policy support, was also specified as continuous throughout the analysis. Similarly, all but the framing variables in our models of distributive fairness—the four social justice principles in addition to perceptions of policies' group impacts—were treated as continuous (Norman, 2010).

All analyses were conducted in RStudio v2024.09.1 using several packages including *sparsereg* for the LASSOplus models (Ratkovic and Tingley, 2017) and *survey* for the descriptive analysis (Lumley, 2004). For all LASSOplus models we used 200 burn-in iterations and 500 Gibbs

sampling iterations. Additionally, to gain confidence in our interpretation of results we supplemented our main models with others. To test the robustness of our distributive fairness and policy support models to policy-specific framing effects, we re-estimated them using a long-format specification with fixed effects for policy type and random effects for respondents. We also conducted ordinal LASSOplus analyses of policy support and distributive fairness, policy-specific ordinary least squares regressions, analysis of variance tests on framing effects, ordinary least squares and LASSOplus models focusing on interactions between distributive fairness and effectiveness variables, and policy-agnostic multi-level and mixed effects models.

4. Results and discussion

4.1. Levels of policy support, and perceptions of distributive fairness and effectiveness

Fig. 1 shows the distribution of policy support levels across the six net-zero instruments included in this study, grouped by message framing treatment. Additionally, 95 % confidence intervals are shown for support (the sum of *support* and *strongly support*) and opposition (the sum of *oppose* and *strongly oppose*). Notably, framing effects are small and there were frequent overlapping confidence intervals. Buildings/Transport Carbon Market, the Electric Appliance Mandate, the Low Carbon Fuel Standard, and the Oil and Gas Cap were supported more than opposed, whereas the opposite was true for the Carbon Tax and ZEV Mandate. Additionally, focusing on the relative proportions of the two opposition categories reveals that in addition to being opposed in general, the proportion of strong opposition was greater for the latter two policies. For example, the ratio of strongly oppose to oppose in the control group

for the Carbon Tax was $21/18 = 1.2$, but was $6/9 = 0.7$ for the Low Carbon Fuel Standard. Relatedly, the higher absolute levels of strong opposition for the Carbon Tax and ZEV Mandate, and to a lesser extent the Electric Appliance Mandate, warn of poor political feasibility (relative to the other policies), especially if these instruments are attached to a high-salience issue, such as the carbon tax in relation to an affordability crisis (Merkley and Owen, 2020).

Fig. 2 explores study participants' perceptions of distributive fairness and effectiveness at reducing emissions across message framing treatments by providing group means and 95 % confidence intervals. On average, respondents believed that the Low Carbon Fuel Standard and Oil and Gas Cap were more effective at reducing emissions and fairer in terms of their perceived impacts than the Buildings/Transport Carbon Market. The latter was rated higher than the ZEV Mandate, which in turn was rated higher than the Carbon Tax. For each policy instrument,

respondents tended to see both policy attributes as matching closely (e. g., for both fairness and effectiveness the Electric Appliance Mandate measures were at approximately a 3 on our 5-point scales), except for the ZEV Mandate, which respondents saw as more effective than fair. Again, in Fig. 2, confidence intervals around average perceptions frequently overlap across framing treatments.

Past studies of Canadian climate policy support consistently found that carbon taxes faced considerably higher opposition than other policy types due to the high salience of their economic costs (Kitt et al., 2021; Lachapelle and Borick, 2022; Long et al., 2020; Mildemberger et al., 2016; Odland et al., 2023; Rhodes et al., 2014, 2017). These findings suggested that regulations, being less visible, might represent the path of least political resistance, allowing governments to impose economic costs for environmental objectives with minimal public pushback. However, our results challenge this assumption. We find that the

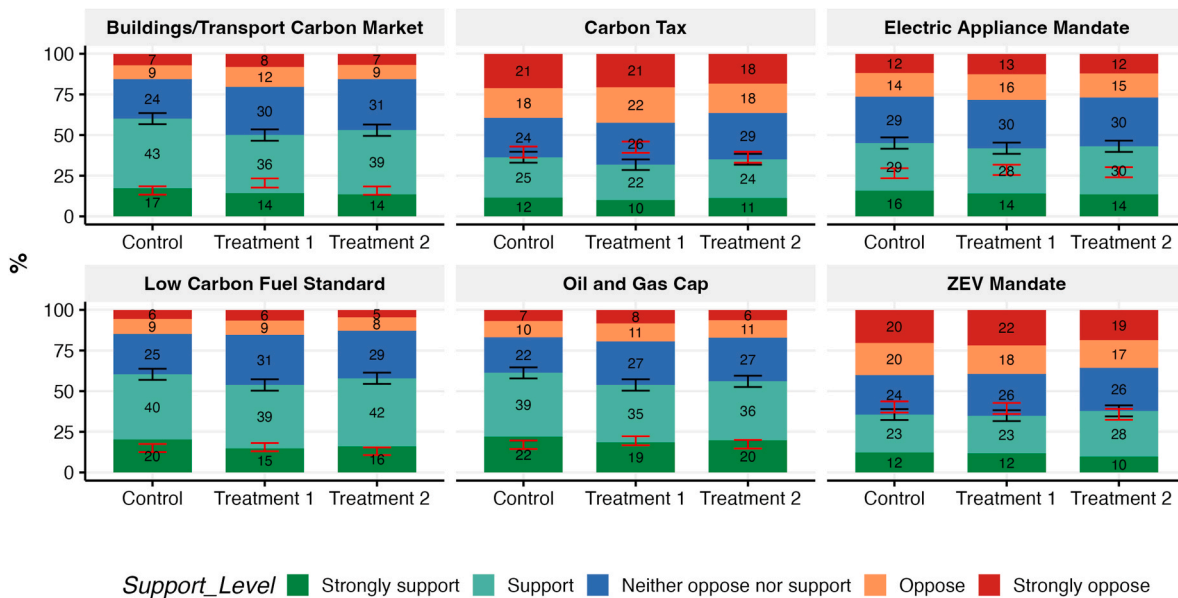


Fig. 1. Distributions of support for climate policies in the national sample by message framing treatment (N = 2359). 95 % confidence intervals are shown in black for support (the sum of *support* and *strongly support*) and in red for opposition (the sum of *oppose* and *strongly oppose*). Percentages may not total 100 due to rounding. The control group received neutral information, Treatment 1 presented a pro-policy *intergenerational fairness* frame along with an anti-policy *economic costs* frame, and Treatment 2 presented a pro-policy *economic benefits* frame along with the same anti-policy *economic costs* frame (see Table 2). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

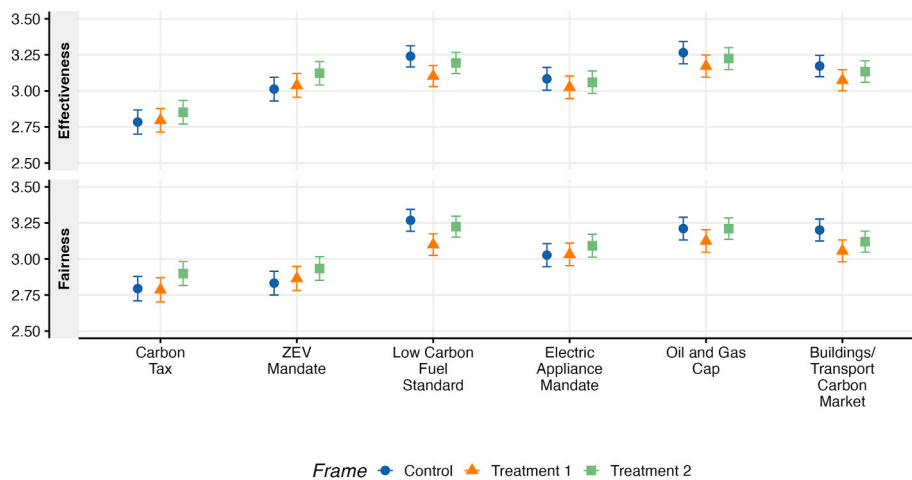


Fig. 2. Perceptions of distributive fairness and effectiveness at reducing emissions (n = 2359). Points represent the mean evaluations, with 95 % confidence intervals shown. The control group received neutral information, Treatment 1 presented a pro-policy *intergenerational fairness* frame along with an anti-policy *economic costs* frame, and Treatment 2 presented a pro-policy *economic benefits* frame along with the same anti-policy *economic costs* frame (see Table 2).

net-zero ZEV Mandate (a regulatory measure) faced levels of opposition comparable to the Carbon Tax, with the Electric Appliance Mandate (also a regulatory measure) close behind. This result is in line with Bell et al., (2025) findings in the United Kingdom and Australia.

Similarly, our findings on perceptions of fairness and effectiveness align with prior research—carbon pricing is typically perceived as less fair and less effective than regulations (Cools et al., 2011; Dreyer et al., 2015; Huber et al., 2020; Thaller et al., 2023). The key exception is the ZEV Mandate, which respondents perceived as just as unfair as the Carbon Tax, yet effective. One plausible explanation is Canadians' strong symbolic attachment to their vehicles (Gauer et al., 2023). However, opposition may also reflect affordability barriers to electric vehicle adoption, infrastructure gaps, or urban-rural differences in feasibility (Long et al., 2025; MacInnis et al., 2023).

Further, a recent global survey of climate policy researchers (Drews et al., 2024) found that direct regulation was rated as the most important instrument in a national policy mix, while carbon taxation and cap-and-trade were rated lower. Although environmental economists expressed greater support for carbon pricing, disciplinary differences were relatively modest. Many experts across fields prioritized the perceived effectiveness and political feasibility of regulatory tools (Drews et al., 2024). These findings align in part with the results presented here: while regulatory policies tended to receive higher public support than carbon pricing, not all regulations were equally popular. This suggests that (1) both expert and public perspectives complicate the common theoretical preference for carbon pricing mechanisms, and (2) while regulation may be broadly preferred, the category itself encompasses a range of instruments that can differ substantially in perceived fairness, effectiveness, and political feasibility.

These findings could imply that as regulatory policies become more stringent over time, their economic costs and coercive nature may become more apparent, leading to heightened public resistance and politicization. However, one advantage of regulatory measures over carbon pricing is that they can be more easily communicated as direct actions to reduce emissions. Unlike carbon pricing, which often requires additional explanation to connect price signals to behavioural change, regulatory policies can be framed as immediate, tangible interventions. Yet, this advantage is not universal—whether a regulation is perceived as an effective and fair climate action or an unacceptable intrusion may depend on its psychological 'distance' from people's everyday experiences (Fiedler et al., 2012; Wang et al., 2019). Policies regulating upstream emissions or industry-wide standards may be seen as abstract and broadly beneficial, reinforcing their association with emission reductions. In contrast, highly visible mandates that explicitly target household technology choices, such as electric vehicle or appliance standards, may trigger resistance by making costs and constraints more salient in everyday life—even if the technological outcomes are very similar to those achieved through inconspicuous approaches.

4.2. Predictors of policy support

Fig. 3 presents the results of six LASSOplus models predicting policy support, one for each net-zero policy instrument. As anticipated, perceived distributive fairness and effectiveness emerged as the strongest predictors of support across all six policies. In line with findings from Bergquist et al. (2022), distributive fairness was generally the stronger predictor, particularly for five of the six instruments. The key exception was the Low Carbon Fuel Standard, where perceived effectiveness had a similar influence on support.

Bergquist et al. (2022) also found that, across their meta-analysis, perceptions of fairness and effectiveness tended to have stronger effects on support for regulatory measures than for economic instruments. Our findings present a slightly different picture. Perceptions of distributive fairness were more influential in shaping support for the Carbon Tax, the Electric Appliance Mandate, and the ZEV Mandate compared to the other three policy instruments. Notably, these three policies also

faced the highest levels of opposition (Fig. 1). In contrast, perceptions of effectiveness had a more consistent influence across all instruments, with no comparable variation in posterior medians.

As discussed earlier, we supplemented this analysis with additional tests to examine potential interactions between perceptions of distributive fairness and effectiveness across the six policy instruments. In theory, fairness perceptions may depend on effectiveness—if people do not believe a policy will achieve its intended goal, they are more likely to perceive it as unfair (Eriksson et al., 2006). However, our results do not support strong interaction effects. The LASSOplus models returned posterior median values of zero for the interaction terms, while ordinary least squares regressions produced insignificant or very small effect sizes. Despite these findings, from a policy perspective, we assume that perceptions of effectiveness can still influence support both directly and indirectly—potentially through their effects on fairness perceptions (Cools et al., 2011).

Message framing treatments are notably absent from Fig. 3 as predictors of support. This is because the posterior medians of the framing treatments are zero in all LASSOplus models of policy support, reflecting the conservative nature of LASSOplus in penalizing weak predictors. This result aligns theoretically with two alternative explanations about strong versus weak frames. According to Chong and Druckman (2007a), strong frames shift opinions, by definition, whereas weak frames fail to do so. One possibility is that respondents perceived both the pro- and anti-policy frames as weak, leading to minimal influence on their policy preferences. Alternatively, both frames may have been perceived as strong, generating opposing considerations that cancelled each other out, resulting in intermediate opinions. The latter seems more likely, given that both the pro- and anti-policy frames used in our study have previously been shown to influence opinions in Canada (see Section 2.4.1).

Regarding covariates, climate change concern was consistently identified as a strong predictor of support across all six policy instruments in the LASSOplus models. Other covariates—including self-transcendent and self-enhancement values, as well as centre-left and centre-right ideology—predicted support for some policy instruments but not others. Notably, trust in implementing institutions and education levels had posterior median values of zero across all six policy instruments, suggesting a limited role in shaping policy support (cf. Fairbrother et al., 2021; Kitt et al., 2021).

4.3. Distributive fairness, perceived impacts, and justice principles

Finally, Fig. 4 presents the results of our exploratory analysis of distributive fairness, examining how perceptions of a policy's impact on different social groups, alongside adherence to distributive justice principles, shaped fairness judgments.² Across all six policy instruments, stronger beliefs in the justice principles of equality and need were negatively associated with perceptions of fairness. Specifically, respondents who more strongly agreed with the following statements were, all else equal, less likely to perceive net-zero policy instruments as fair: (a) "A society is fair when income and wealth are equally distributed among all people," and (b) "A society is fair when it takes care of those who are poor and in need, regardless of what they give back." This finding highlights a potential challenge: while ideologically left-leaning respondents were more likely to view climate change as a political priority—mirroring national trends (Lachapelle and Borick, 2022)—they were less likely to perceive stringent policies as fair.

Perceptions of distributive fairness were most strongly predicted by

² To assess the robustness of our findings to unobserved, policy-specific characteristics (e.g., framing effects or embedded cost cues), we re-estimated the distributive fairness model in a pooled specification with fixed effects for policy type and respondent-level random effects. Key predictors remained credibly associated with fairness outcomes.

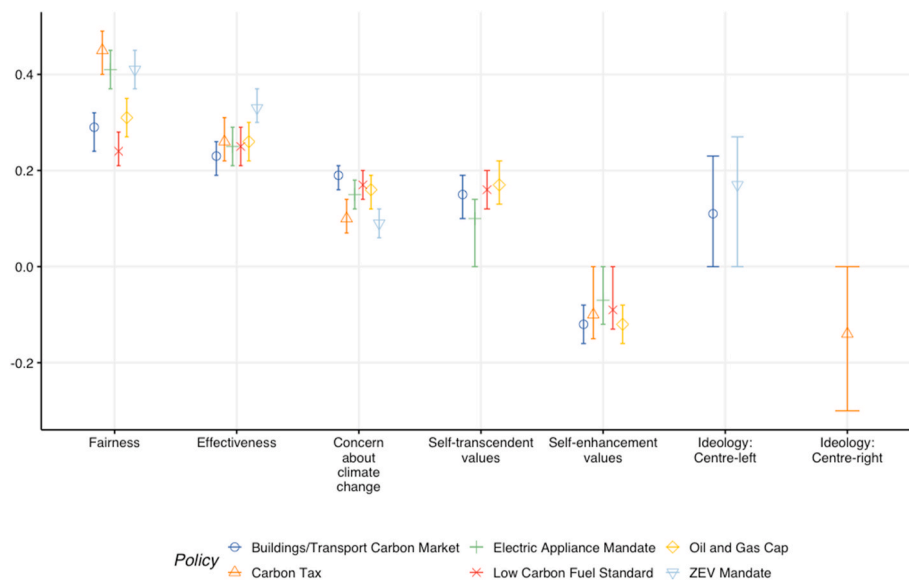


Fig. 3. Results of LASSOplus models of policy support across six policy instruments (N = 2312). All non-zero posterior medians are shown along with 90 % credible intervals.

three social groups: future generations of Canadians, people with low incomes, and residents of rural areas or small towns. In contrast, other groups—such as small business owners, urban residents, and fossil fuel workers—show mixed or weaker predictive relationships. This finding aligns with Lindvall et al. (2024), who suggested that public support for carbon taxes in Sweden can be increased by distributing revenues to low-income earners and rural residents. However, Canada's experience with carbon pricing suggests that a progressive dividend system has had only a minimal effect on policy support (Dechezleprêtre et al., 2022). Meanwhile, perceptions of fairness appeared unaffected by perceptions of policy impacts on groups such as clean energy workers, high-income earners, and oil and gas shareholders, as indicated by their posterior medians of zero.

4.4. Limitations

This study has several limitations. First, as a cross-sectional survey, the findings capture a single point in time and cannot account systematically for how public opinion evolves. Longitudinal research is needed to track stability and change in policy support over time. Relatedly, while message framing can shape policy perceptions, its effects are context dependent. Our study does not control for frequency of exposure to frames, which plays a crucial role in real-world persuasion (Druckman et al., 2010), and survey participants are guaranteed exposure, unlike in real-world political communication, where audiences selectively engage with information (Kinder, 2007). Moreover, our frames are self-reported 'strong' frames, but we did not confirm this with external validation. Also, textual framing in a survey experiment does not replicate the broader media environment or the persuasive influence of political elites and visual or affective cues.

Additionally, our survey responses are susceptible to hypothetical bias, where participants may express different preferences than they would in real-world settings (i.e., stated and actual behaviour). Also, the study identifies associations rather than causal relationships. While our conceptual model aligns with prior research, we did not systematically test interactions among key explanatory variables, such as demographic subgroups or psychological constructs, which may moderate policy support. Our measure of distributive fairness relies on the Basic Social Justice Orientation scale (Van Hootegeem et al., 2021), which, while widely used, provides indicators that focus on specific elements of distributive justice but may not fully capture the complexity of fairness

judgments. Alternative measures, particularly those incorporating latent constructs, could enhance future research on fairness perceptions.

Strong opposition to the ZEV Mandate and, to a lesser extent, the Electric Appliance Mandate may reflect that the 'policy information' provided to respondents explicitly highlighted direct cost implications for consumers, making them tangible 'kitchen table' issues. In contrast, measures such as the Low Carbon Fuel Standard, Oil and Gas Cap, or the Buildings and Transport GHG Permits may not have clearly communicated direct personal cost implications, potentially explaining their lower opposition levels. Future research could experimentally test the effect of explicitly highlighting or obscuring consumer cost impacts in policy messaging.

Next, while our model of distributive fairness considers perceived policy impacts on social groups, it is constrained by the list of groups we selected. The model is useful for simply comparing fairness perceptions across these groups, but it does not capture fluid or intersectional identities that may influence policy attitudes in nuanced ways. Future research could complement this approach with open-ended questions and qualitative analysis to capture more nuanced perceptions of fairness, particularly where the number or type of social groups affected by climate policies may exceed predefined categories (see e.g., Savin et al., 2022).

Further, although this study emphasizes distributive fairness, procedural fairness is also a critical component of public acceptability (Bergquist et al., 2022; Tyler, 2000). Procedural dimensions such as transparency, accountability, and participation in decision-making may independently or interactively influence support for climate policies. Future research could incorporate measures of procedural fairness to better untangle their relative contributions to policy support.

Finally, differences in political culture, governance structures, and economic conditions mean that generalizing to national contexts other than Canada must be done cautiously. Countries with distinct institutional trust levels, varying levels of economic inequality, or different histories of climate policy implementation and economic development may yield different results regarding public support for net-zero policies. Future research could test the replicability of these dynamics in other national settings.

5. Conclusions and policy implications

This study contributes to the literature on public support for climate

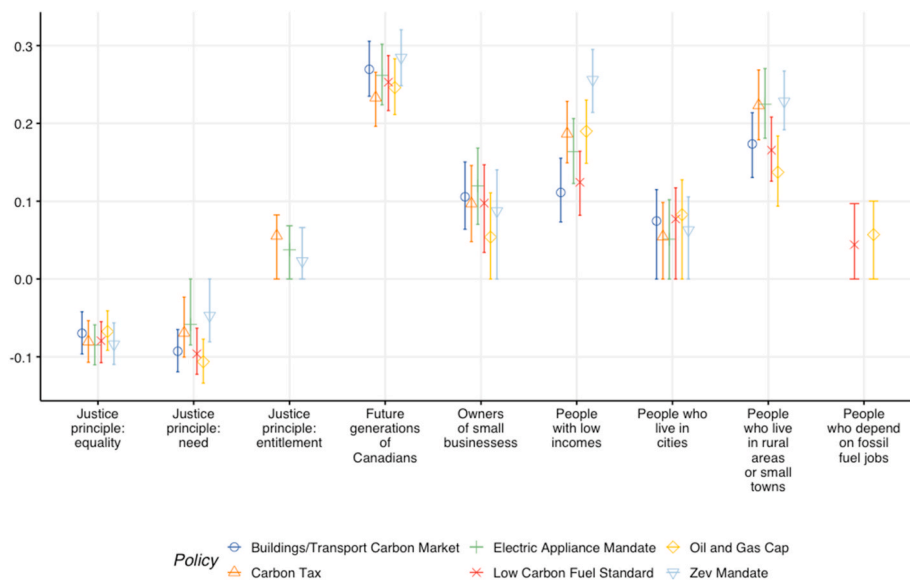


Fig. 4. Results of LASSOplus models of distributive fairness (N = 2312). All non-zero posterior medians are shown along with 90 % credible intervals.

policies by examining how perceptions of distributive fairness and effectiveness influence support for net-zero policy instruments in Canada. By incorporating policy-specific attributes alongside individual and contextual factors, our findings refine the existing understanding of the drivers of policy support. The study explores the influence of two pro-policy message frames when positioned against a competing anti-policy message. These findings can be considered by policymakers seeking to implement durable and politically viable climate policies.

First, in this particular context, the competing frames tested did not shift public support when presented together. This suggests that real-world debates, where competing frames often coexist, may neutralize the persuasive power of individual frames (Aklin and Urpelainen, 2013). The anti-policy frame emphasizing the economic cost concerns of policy implementation was likely persuasive due to heightened affordability concerns in Canada at the time of the survey (August 2024). Notably, prior studies using one-sided experiments have found similar pro-policy frames to be persuasive, suggesting that the ‘non-effect’ in this study stems from competition rather than an inherent weakness in the pro-policy messaging. This finding aligns with the evidence suggesting policymakers should be sensitive to issue salience when communicating policy information (Ciuk and Yost, 2016).

Our results reinforce the significance of distributive fairness in shaping public support for climate policies. As in prior research, perceptions of fairness emerge as the strongest predictor of support, exceeding perceptions of effectiveness for most policy instruments (Bergquist et al., 2022; Thaller et al., 2023). We also show, across different policy instruments, how distributive fairness judgments are influenced by individuals’ adherence to different distributive justice principles and their perceptions of a policy’s impact on specific social groups. Specifically, we find that respondents who place a greater emphasis on equality- and need-based justice principles are less likely to perceive net-zero policies as fair, potentially reflecting concerns about affordability and equity in policy design. Additionally, perceptions of fairness are most strongly associated with anticipated policy impacts on future generations, low-income households, and rural communities—suggesting that distributive fairness concerns are driven by both intergenerational and socio-economic considerations.

Perceived effectiveness also remains a strong predictor of support, particularly for policies framed as directly reducing GHG emissions. While fairness and effectiveness are conceptually linked—respondents who see a policy as ineffective are also less likely to see it as fair—our analysis does not find strong interaction effects between the two. This

suggests that while perceptions of distributional fairness and effectiveness often align, they may operate as distinct but complementary drivers of support rather than reinforcing one another in a systematic way. Even so, communicators should consider both in tandem, as messaging around effectiveness may still indirectly influence fairness perceptions and, by extension, public acceptance. However, simply asserting a policy’s effectiveness is insufficient; credibility remains key. Trusted institutions (e.g., independent regulatory bodies) or non-partisan experts are usually better positioned than politicians to communicate effectiveness claims persuasively (Kuntze and Fesenfeld, 2021).

Policy instrument selection plays a crucial role in shaping public perceptions. While past research suggests that command-and-control regulations are generally better supported and perceived as fairer and more effective than market-based instruments such as carbon pricing, our findings complicate this narrative. We observe that opposition to a mandate, and to a lesser extent an electric appliance mandate, rivals that of a consumer carbon tax, despite their regulatory nature. Moreover, fairness perceptions of the ZEV mandate are particularly low, on par with the carbon tax, even as it is viewed as effective.

These findings suggest that policy adoption could benefit from communication and design strategies that emphasize both distributive fairness and perceived effectiveness. Future research could examine how these perceptions vary across political contexts, evolve in response to public deliberation or lived experience, and interact with procedural engagement mechanisms such as citizen assemblies or participatory policy design.

Regarding communication, policymakers aiming to build support for net-zero transitions may benefit from communication strategies that foreground fairness considerations. Our findings suggest that emphasizing how policies benefit disadvantaged groups (e.g., low-income households, rural residents, future generations) may enhance perceptions of distributive fairness and, in turn, public support. Messaging that links policy goals to widely held values such as equity, responsibility, and care for others, may also improve receptivity by resonating with core justice principles. Clear articulation of who benefits, why, and how trade-offs are addressed may help to pre-empt misperceptions and strengthen perceived legitimacy.

Beyond initial adoption, climate policies must be designed for long-term durability. Our findings also suggest that policy feedback mechanisms, where policies themselves generate public support over time, may be critical. Policies that provide visible and substantive benefits to key constituencies can help build coalitions of support, or perhaps more

crucially, mitigate opposition from those who might otherwise resist them. Conversely, policies perceived as unfair or ineffective risk generating negative feedback. Where possible, decision-makers should consider a holistic approach to policy design by aligning credible effectiveness claims with strategic communication while anticipating and addressing public perceptions of distributional policy outcomes.

CRedit authorship contribution statement

Aaron Hoyle: Writing – original draft, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Ekaterina Rhodes:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization, Supervision.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enpol.2025.114644>.

Data availability

Data will be made available on request.

References

- Adriaans, J., Fourré, M., 2022. Basic social justice orientations—Measuring order-related justice in the European social survey round 9. *Measur. Instrum. Soc. Sci.* 4 (1), 11. <https://doi.org/10.1186/s42409-022-00040-3>.
- Aklin, M., Urpelainen, J., 2013. Debating clean energy: frames, counter frames, and audiences. *Glob. Environ. Change* 23 (5), 1225–1232. <https://doi.org/10.1016/j.gloenvcha.2013.03.007>.
- Ansolabehere, S., Konisky, D.M., 2014. *Cheap and Clean: How Americans Think About Energy in the Age of Global Warming*. The MIT Press. <https://doi.org/10.7551/mitpress/9999.001.0001>.
- Badulovich, N., Grant, W.J., Colvin, R.M., 2020. Framing climate change for effective communication: a systematic map. *Environ. Res. Lett.* 15 (12), 123002. <https://doi.org/10.1088/1748-9326/aba4c7>.
- Bayulgen, O., Benegal, S., 2019. Green priorities: how economic frames affect perceptions of renewable energy in the United States. *Energy Res. Social Sci.* 47, 28–36. <https://doi.org/10.1016/j.erss.2018.08.017>.
- Bell, C., Rhodes, E., Long, Z., Salemi, C., 2025. Do economic trade-offs matter in climate policy support? Survey evidence from the United Kingdom and Australia. *Energy Policy* 197, 114430. <https://doi.org/10.1016/j.enpol.2024.114430>.
- Bergquist, M., Nilsson, A., Harring, N., Jagers, S.C., 2022. Meta-analyses of fifteen determinants of public opinion about climate change taxes and laws. *Nat. Clim. Change* 12 (3). <https://doi.org/10.1038/s41558-022-01297-6>. Article 3.
- Bumann, S., 2021. What are the determinants of public support for climate policies? A review of the empirical literature. *Rev. Econ.* 72 (3), 213–228. <https://doi.org/10.1515/roe-2021-0046>.
- Canadian Institute for Climate Choices, 2021. Canada's net zero future: finding our way in the global transition. <https://climatechoices.ca/wp-content/uploads/2021/02/Canas-Net-Zero-Future-FINAL-2.pdf>.
- Chong, D., Druckman, J.N., 2007a. A theory of framing and opinion formation in competitive elite environments. *J. Commun.* 57 (1), 99–118. <https://doi.org/10.1111/j.1460-2466.2006.00331.x>.
- Chong, D., Druckman, J.N., 2007b. Framing theory. *Annu. Rev. Polit. Sci.* 10 (1), 103–126. <https://doi.org/10.1146/annurev.polisci.10.072805.103054>.
- Chryst, B., Marlon, J., Van Der Linden, S., Leiserowitz, A., Maibach, E., Roser-Renouf, C., 2018. Global warming's "six Americas short survey": audience segmentation of climate change views using a four question instrument. *Environ. Commun.* 12 (8), 1109–1122. <https://doi.org/10.1080/17524032.2018.1508047>.
- Ciuk, D.J., Yost, B.A., 2016. The effects of issue salience, elite influence, and policy content on public opinion. *Polit. Commun.* 33 (2), 328–345. <https://doi.org/10.1080/10584609.2015.1017629>.
- Clark, C.F., Kotchen, M.J., Moore, M.R., 2003. Internal and external influences on pro-environmental behavior: participation in a green electricity program. *J. Environ. Psychol.* 23 (3), 237–246. [https://doi.org/10.1016/S0272-4944\(02\)00105-6](https://doi.org/10.1016/S0272-4944(02)00105-6).
- Re.Climate, 2024. What do Canadians really think about climate change? - a summary of public opinion research for communicators. <https://reclimate.ca/wp-content/uploads/2024/03/2024-public-opinion-summary.pdf>.
- Cools, M., Brijis, K., Tormans, H., Moons, E., Janssens, D., Wets, G., 2011. The socio-cognitive links between road pricing acceptability and changes in travel-behavior. *Transport. Res. Pol. Pract.* 45 (8), 779–788. <https://doi.org/10.1016/j.tra.2011.06.006>.
- Czarnek, G., Kossowska, M., Szwed, P., 2021. Right-wing ideology reduces the effects of education on climate change beliefs in more developed countries. *Nat. Clim. Change* 11 (1), 9–13. <https://doi.org/10.1038/s41558-020-00930-6>.
- Dasandi, N., Graham, H., Hudson, D., Jankin, S., vanHeerde-Hudson, J., Watts, N., 2022. Positive, global, and health or environment framing bolsters public support for climate policies. *Commun. Earth Environ.* 3 (1), 239. <https://doi.org/10.1038/s43247-022-00571-x>.
- Dechezleprêtre, A., Fabre, A., Kruse, T., Planterose, B., Chico, A.S., Stantcheva, S., 2022. *Fighting Climate Change: International Attitudes Toward Climate Policies* (No. w30265). National Bureau of Economic Research.
- Dietz, T., Fitzgerald, A., Shwom, R., 2005. Environmental values. *Annu. Rev. Environ. Resour.* 30, 335–372. <https://doi.org/10.1146/annurev.energy.30.050504.144444>. Volume 30, 2005.
- Dietz, T., Dan, A., Shwom, R., 2007. Support for climate change policy: social psychological and social structural influences. *Rural Sociol.* 72 (2), 185–214. <https://doi.org/10.1526/003601107781170026>.
- Dillman, D.A., Smyth, J.D., Christian, L.M., 2014. *Internet, Phone, Mail, and mixed-mode Surveys: the Tailored Design Method*, fourth ed. Wiley.
- Drews, S., van den Bergh, J.C.J.M., 2016. What explains public support for climate policies? A review of empirical and experimental studies. *Clim. Policy* 16 (7), 855–876. <https://doi.org/10.1080/14693062.2015.1058240>.
- Drews, S., Savin, I., van den Bergh, J.C.J.M., 2022. Biased perceptions of other people's attitudes to carbon taxation. *Energy Policy* 167, 113051. <https://doi.org/10.1016/j.enpol.2022.113051>.
- Drews, S., Savin, I., van den Bergh, J., 2024. A global survey of scientific consensus and controversy on instruments of climate policy. *Ecol. Econ.* 218, 108098. <https://doi.org/10.1016/j.ecolecon.2023.108098>.
- Dreyer, S.J., Walker, I., McCoy, S.K., Teisl, M.F., 2015. Australians' views on carbon pricing before and after the 2013 federal election. *Nature Climate Change* 5 (12), 1064–1067. <https://doi.org/10.1038/nclimate2756>.
- Druckman, J.N., 2001. Evaluating framing effects. *J. Econ. Psychol.* 22 (1), 91–101. [https://doi.org/10.1016/S0167-4870\(00\)00032-5](https://doi.org/10.1016/S0167-4870(00)00032-5).
- Druckman, J.N., Hennessy, C.L., St Charles, K., Webber, J., 2010. Competing rhetoric over time: frames versus cues. *J. Polit.* 72 (1), 136–148. <https://doi.org/10.1017/S0022381609990521>.
- Dua, R., 2024. Net-zero transport dialogue: emerging developments and the puzzles they present. *Energy Sustain. Dev.* 82, 101516. <https://doi.org/10.1016/j.esd.2024.101516>.
- Environment and Climate Change Canada, 2022. Exploring approaches for Canada's transition to net-zero emissions: Canada's long-term strategy submission to the United Nations framework convention on climate change. https://unfccc.int/sites/default/files/resource/LTS%20Full%20Draft_Final%20version_oct31.pdf.
- Environment and Climate Change Canada, 2024. National inventory report, 1990–2022: greenhouse gas sources and sinks in Canada. *Environment and Climate Change Canada*. climatechange.ca/ghg-inventory.
- Eriksson, L., Garvill, J., Nordlund, A.M., 2006. Acceptability of travel demand management measures: the importance of problem awareness, personal norm, freedom, and fairness. *J. Environ. Psychol.* 26 (1), 15–26. <https://doi.org/10.1016/j.jenvp.2006.05.003>.
- Fairbrother, M., Arrhenius, G., Bykvist, K., Campbell, T., 2021. Governing for future generations: how political trust shapes attitudes towards climate and debt policies. *Front. Political Sci.* 3. <https://doi.org/10.3389/fpos.2021.656053>.
- Fesenfeld, L., Beiser-McGrath, L., Sun, Y., Wicki, M., Bernauer, T., 2024. Systematic mapping of climate and environmental framing experiments and re-analysis with computational methods points to omitted interaction bias. *PLOS Climate* 3 (2), e0000297. <https://doi.org/10.1371/journal.pclm.0000297>.
- Fiedler, K., Jung, J., Wänke, M., Alexopoulos, T., 2012. On the relations between distinct aspects of psychological distance: an ecological basis of construal-level theory. *J. Exp. Soc. Psychol.* 48 (5), 1014–1021. <https://doi.org/10.1016/j.jesp.2012.03.013>.
- Gauer, V.H., Axsen, J., Dütschke, E., Long, Z., Kelber, A., 2023. Who is more attached to their car? Comparing automobility engagement and response to shared, automated and electric mobility in Canada and Germany. *Energy Research & Social Science* 99, 103048. <https://doi.org/10.1016/j.erss.2023.103048>.
- Grelle, S., Hofmann, W., 2024. When and why do people accept public-policy interventions? An integrative public-policy-acceptance framework. *Perspect. Psychol. Sci.* 19 (1), 258–279. <https://doi.org/10.1177/17456916231180580>.
- Harring, N., Jagers, S., Matti, S., 2017. Public support for pro-environmental policy measures: examining the impact of personal values and ideology. *Sustainability* 9 (5), 679. <https://doi.org/10.3390/su9050679>.
- Harrison, K., 2023. Climate governance and federalism in Canada. In: Fenna, A., Jodoin, S., Setzer, J. (Eds.), *Climate Governance and Federalism*, first ed. Cambridge University Press, pp. 64–85. <https://doi.org/10.1017/9781009249676.005>.
- Heyen, D.A., 2023. Social justice in the context of climate policy: systematizing the variety of inequality dimensions, social impacts, and justice principles. *Clim. Policy* 23 (5), 539–554. <https://doi.org/10.1080/14693062.2022.2142499>.

- Heyen, D.A., Wicki, M., 2024. Increasing public support for climate policy proposals: a research agenda on governable acceptability factors. *Clim. Policy* 24 (6), 785–794. <https://doi.org/10.1080/14693062.2024.2330390>.
- Howlett, M., 2019. *Designing Public Policies: Principles and Instruments*, second ed. Routledge, Taylor & Francis Group.
- Hoyle, A., Peters, J., Jaccard, M., Rhodes, E., 2024. Additional or accidental? Simulating interactions between a low-carbon fuel standard and other climate policy instruments in Canada. *Energy Policy* 185, 113919. <https://doi.org/10.1016/j.enpol.2023.113919>.
- Huber, L., 2022. Beyond policy: the use of social group appeals in party communication. *Polit. Commun.* 39 (3), 293–310. <https://doi.org/10.1080/10584609.2021.1998264>.
- Huber, L.M., Haselmayer, M., 2024. Promising links: how parties combine policy issues with group appeals. *West Eur. Polit.* <https://www.tandfonline.com/doi/abs/10.1080/01402382.2024.2351345>.
- Huber, R.A., Wicki, M.L., Bernauer, T., 2020. Public support for environmental policy depends on beliefs concerning effectiveness, intrusiveness, and fairness. *Environ. Polit.* 29 (4), 649–673. <https://doi.org/10.1080/09644016.2019.1629171>.
- Kahneman, D., 2011. *Thinking, Fast and Slow*, first ed. Farrar, Straus and Giroux.
- Kallbekken, S., 2023. Research on public support for climate policy instruments must broaden its scope. *Nat. Clim. Change* 1–3. <https://doi.org/10.1038/s41558-022-01593-1>.
- Kinder, D.R., 2007. Curmudgeonly advice. *J. Commun.* 57 (1), 155–162. <https://doi.org/10.1111/j.1460-2466.2006.00335.x>.
- Kitt, S., Axsen, J., Long, Z., Rhodes, E., 2021. The role of trust in citizen acceptance of climate policy: comparing perceptions of government competence, integrity and value similarity. *Ecol. Econ.* 183, 106958. <https://doi.org/10.1016/j.ecolecon.2021.106958>.
- Klinsky, S., Dowlatabadi, H., 2009. Conceptualizations of justice in climate policy. *Clim. Policy* 9 (1), 88–108. <https://doi.org/10.3763/cpol.2007.0468>.
- Konc, T., Drews, S., Savin, I., van den Bergh, J.C.J.M., 2022. Co-dynamics of climate policy stringency and public support. *Glob. Environ. Change* 74, 102528. <https://doi.org/10.1016/j.gloenvcha.2022.102528>.
- Kuntze, L., Pesenfeld, L.P., 2021. Citizen assemblies can enhance political feasibility of ambitious climate policies. *Social Science Research Network*. <https://doi.org/10.2139/ssrn.3918532> (SSRN Scholarly Paper 3918532).
- Lachapelle, E., Borick, C., 2022. A Decade of Comparative Canadian and American Public Opinion on Climate Change. *North American Colloquium*. <https://fordschool.umich.edu/research/north-american-colloquium-climate-series/decade-comparative-canadian-and-american-public>.
- Li, N., Su, L.Y.-F., 2018. Message framing and climate change communication: a meta-analytical review. *J. Appl. Commun.* 102 (3). <https://doi.org/10.4148/1051-0834.2189>.
- Lindvall, D., Sörqvist, P., Jagers, S.C., Karlsson, M., Sjöberg, S., Barthel, S., 2024. The role of fairness for accepting stricter carbon taxes in Sweden. *Climate* 12 (11), 11. <https://doi.org/10.3390/cli12110170>.
- Lokshin, M.M., Torre, I., Hannon, M., Purroy, M., 2024. Do More Informed Citizens Make Better Climate Policy Decisions? *World Bank*. <https://hdl.handle.net/10986/42195>.
- Long, Z., Axsen, J., Kitt, S., 2020. Public support for supply-focused transport policies: vehicle emissions, low-carbon fuels, and ZEV sales standards in Canada and California. *Transport. Res. Pol. Pract.* 141, 98–115. <https://doi.org/10.1016/j.tra.2020.08.008>.
- Long, Z., Axsen, J., Gauer, V.H., Niet, T., 2025. Differences in Canadian consumers' awareness and preferences for zero-emissions vehicles from 2013 to 2023. *Transport. Res. Transport Environ.* 141, 104633. <https://doi.org/10.1016/j.trd.2025.104633>.
- Lumley, T., 2004. Analysis of complex survey samples. *J. Stat. Software* 9, 1–19. <https://doi.org/10.18637/jss.v009.i08>.
- MacInnis, B., Krosnick, J.A., McDonald, J., 2023. Resistance to purchasing all-electric vehicles: evidence from a national survey. *J. Environ. Psychol.* 91, 102114. <https://doi.org/10.1016/j.jenvp.2023.102114>.
- Maestre-Andrés, S., Drews, S., van den Bergh, J., 2019. Perceived fairness and public acceptability of carbon pricing: a review of the literature. *Clim. Policy* 19 (9), 1186–1204. <https://doi.org/10.1080/14693062.2019.1639490>.
- Maestre-Andrés, S., Drews, S., Savin, I., van den Bergh, J., 2021. Carbon tax acceptability with information provision and mixed revenue uses. *Nat. Commun.* 12 (1). <https://doi.org/10.1038/s41467-021-27380-8>, 7017–7017.
- Martel-Morin, M., Lachapelle, E., 2022. The five canadas of climate change: using audience segmentation to inform communication on climate policy. *PLoS One* 17 (11), e0273977. <https://doi.org/10.1371/journal.pone.0273977>.
- Merkley, E., Owen, A., 2020. Back to the future: democratic responsiveness and the estimation of future public opinion. *Int. J. Publ. Opin. Res.* 32 (2), 203–222. <https://doi.org/10.1093/ijpor/edz223>.
- Mildenberger, M., Howe, P., Lachapelle, E., Stokes, L., Marlon, J., Gravelle, T., 2016. The distribution of climate change public opinion in Canada. *PLoS One* 11 (8). <https://doi.org/10.1371/journal.pone.0159774>.
- Millar, H., Bourgeois, E., Bernstein, S., Hoffmann, M., 2021. Self-reinforcing and self-undermining feedbacks in subnational climate policy implementation. *Environ. Polit.* 30 (5), 791–810. <https://doi.org/10.1080/09644016.2020.1825302>.
- Navius Research, 2023. Net zero Canada methodology report: Energy-economy modelling of five net zero pathways for clean prosperity. https://www.naviusresearch.com/publications/net_zero_pathways.
- Nelson, T.E., Kinder, D.R., 1996. Issue frames and group-centrism in American public opinion. *J. Polit.* 58 (4), 1055–1078. <https://doi.org/10.2307/2960149>.
- Norman, G., 2010. Likert scales, levels of measurement and the “laws” of statistics. *Adv. Health Sci. Educ. : Theor. Pract.* 15 (5), 625–632. <https://doi.org/10.1007/s10459-010-9222-y>.
- Odland, S., Rhodes, E., Corbett, M., Pardy, A., 2023. What policies do homeowners prefer for building decarbonization and why? An exploration of climate policy support in Canada. *Energy Policy* 173, 113368. <https://doi.org/10.1016/j.enpol.2022.113368>.
- Poortinga, W., Whitmarsh, L., Steentjes, K., 2023. Factors and framing effects in support for net zero policies in the United Kingdom. *Front. Psychol.* 14. <https://doi.org/10.3389/fpsyg.2023.1287188>.
- Potential Energy, 2023. Later is too late. https://potentialenergycoalition.org/wp-content/uploads/2024/01/Later_is_Too_Late_Global_Report.pdf.
- Ratkovic, M., Tingley, D., 2017. Sparse estimation and uncertainty with application to subgroup analysis. *Polit. Anal.* 25 (1), 1–40. <https://doi.org/10.1017/pan.2016.14>.
- Rhodes, E., Axsen, J., Jaccard, M., 2014. Does effective climate policy require well-informed citizen support? *Glob. Environ. Change* 29, 92–104. <https://doi.org/10.1016/j.gloenvcha.2014.09.001>.
- Rhodes, E., Axsen, J., Jaccard, M., 2017. Exploring citizen support for different types of climate policy. *Ecol. Econ.* 137, 56–69. <https://doi.org/10.1016/j.ecolecon.2017.02.027>.
- Rosenbloom, D., 2018. Framing low-carbon pathways: a discursive analysis of contending storylines surrounding the phase-out of coal-fired power in Ontario. *Environ. Innov. Soc. Transit.* 27, 129–145. <https://doi.org/10.1016/j.eist.2017.11.003>.
- Savin, I., Drews, S., van den Bergh, J., Villamayor-Tomas, S., 2022. Public expectations about the impact of COVID-19 on climate action by citizens and government. *PLoS One* 17 (6), e0266979. <https://doi.org/10.1371/journal.pone.0266979>.
- Simon, H.A., 1955. A behavioral model of rational choice. *Q. J. Econ.* 69, 99–118. <https://doi.org/10.2307/1884852>.
- Stoddart, M.C.J., Haluza-Delay, R., Tindall, D.B., 2016. Canadian news media coverage of climate change: historical trajectories, dominant frames, and international comparisons. *Soc. Nat. Resour.* 29 (2), 218–232. <https://doi.org/10.1080/08941920.2015.1054569>.
- Tajfel, H., Turner, J., 2001. *An integrative theory of intergroup conflict*. In: *Intergroup Relations: Essential Readings*. Psychology Press, pp. 94–109.
- Tajfel, H., Turner, J.C., 2004. *The Social Identity Theory of Intergroup Behavior*. Psychology Press, p. 293. <https://doi.org/10.4324/9780203505984-16>.
- Thaller, A., Fleiß, E., Brohmer, H., Köstenbaumer, D., Posch, A., Athenstaedt, U., 2023. When perceived fairness and acceptance go hand in hand—drivers of regulatory and economic policies for low-carbon mobility. *PLOS Climate* 2, e0000157. <https://doi.org/10.1371/journal.pclm.0000157>.
- Thomas, M., DeCillia, B., Santos, J.B., Thorlakson, L., 2022. Great expectations: public opinion about energy transition. *Energy Policy* 162, 112777. <https://doi.org/10.1016/j.enpol.2022.112777>.
- Tyler, T.R., 2000. Social justice: outcome and procedure. *Int. J. Psychol.* 35 (2), 117–125. <https://doi.org/10.1080/002075900399411>.
- Tyler, T.R., Lind, E.A., 1992. *The social psychology of procedural justice*. *Adv. Exp. Soc. Psychol.* 25, 137.
- Van Hootegeem, A., Meuleman, B., Abts, K., 2021. Measuring public support for distributive justice principles: assessing the measurement quality of the basic social justice orientations scale. *Int. J. Publ. Opin. Res.* 33 (4), 986–997. <https://doi.org/10.1093/ijpor/edaa041>.
- Wang, S., Hurlstone, M.J., Leviston, Z., Walker, I., Lawrence, C., 2019. Climate change from a distance: an analysis of construal level and psychological distance from climate change. *Front. Psychol.* 10, 230. <https://doi.org/10.3389/fpsyg.2019.00230>.